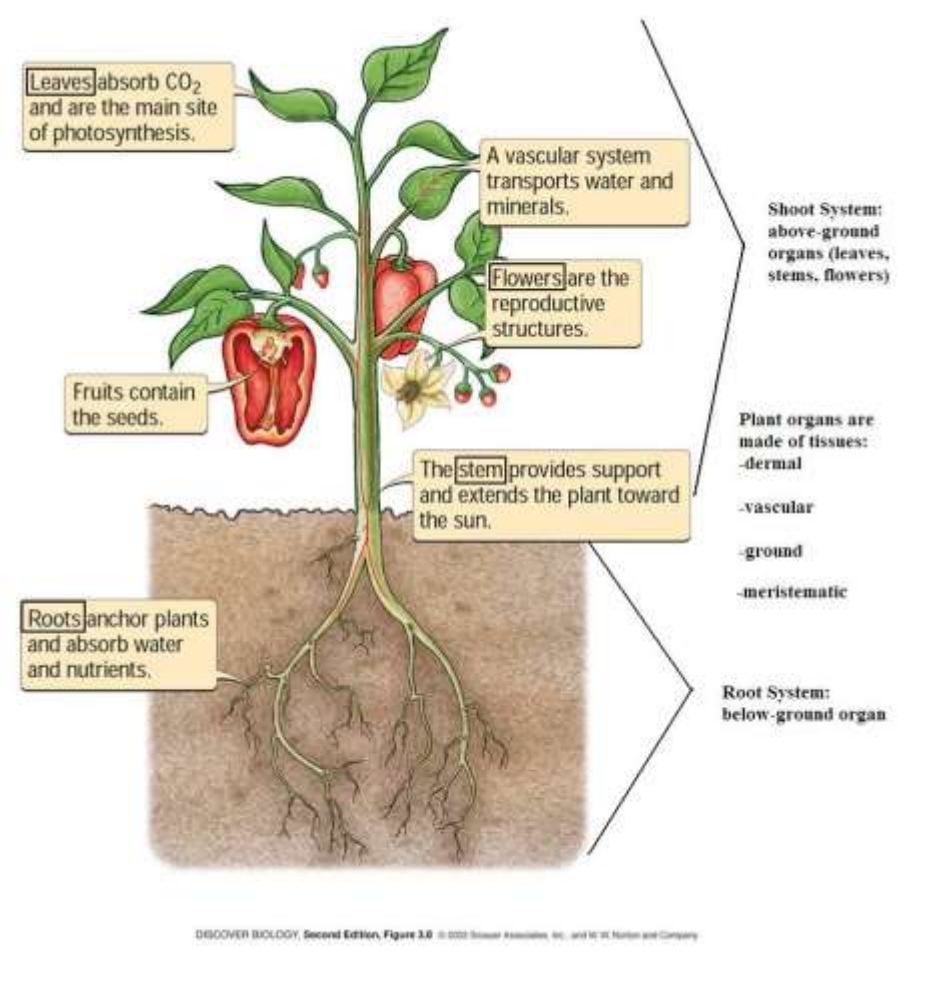
Plant Anatomy

Plant Organs

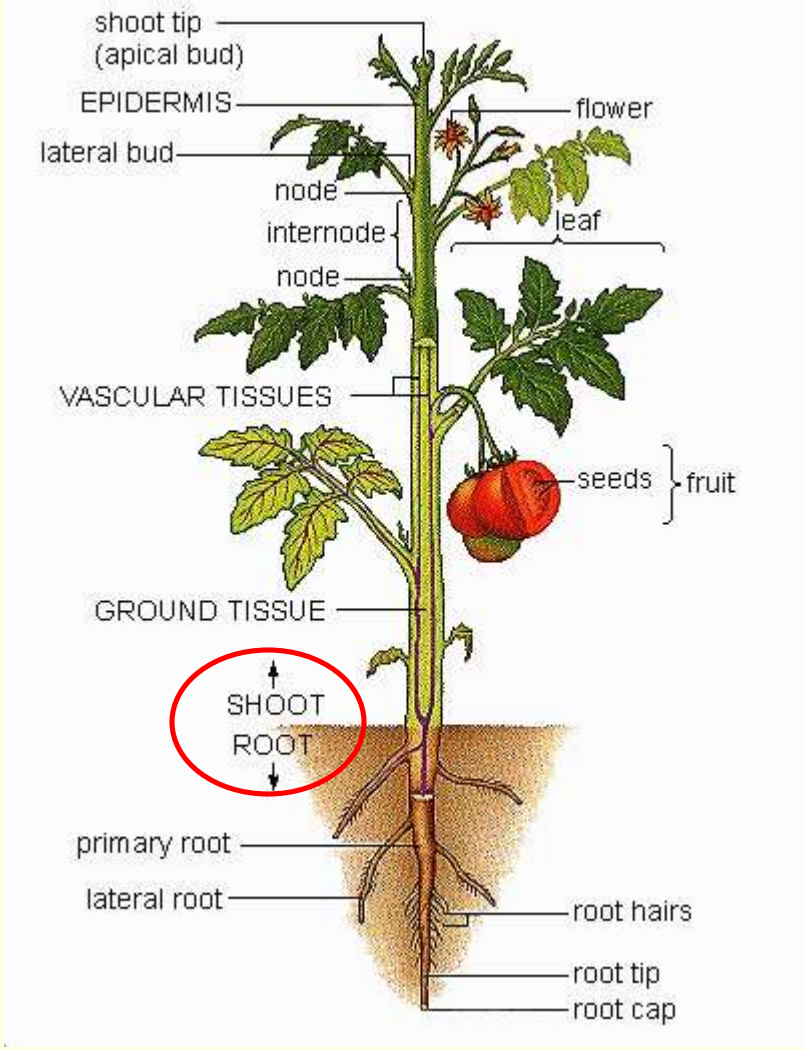
Types of Plant Organs

- Vegetative organs:
  - Roots
  - Leaves
  - Stems
- Reproductive organs:
  - Flowers
  - Fruit



Plant Body Systems

- The plant body is organized into a root system and a shoot system:
  - Root system is generally below ground.
  - Shoot system consists of vertical stems, leaves, flowers, & fruit that contain seeds.

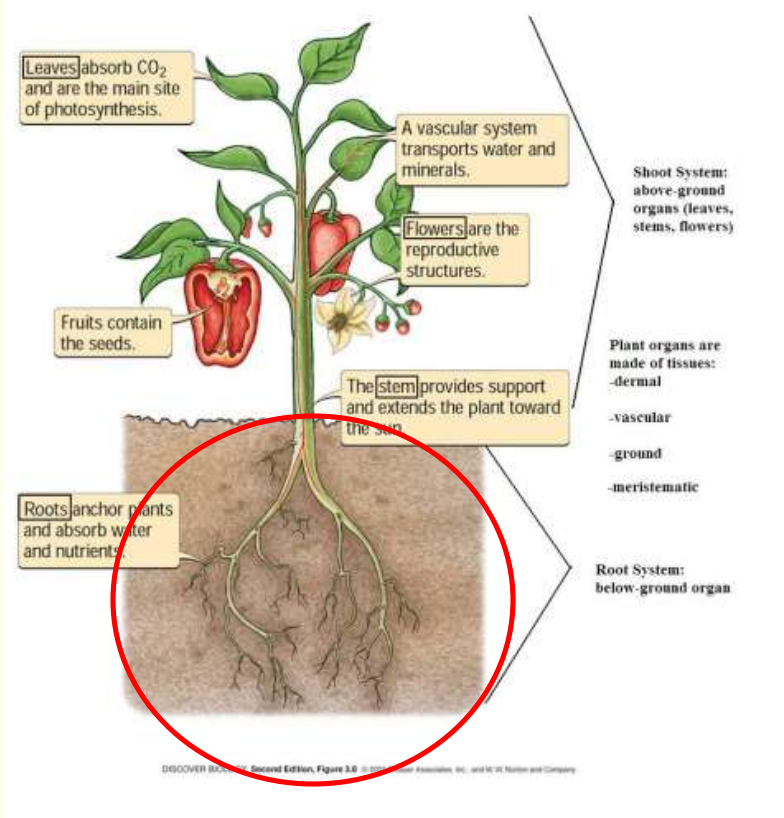


Root System

A plant’s “sponge” and “anchor” …

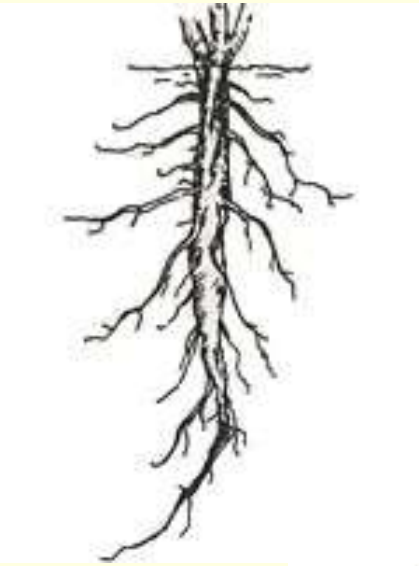
The Root System

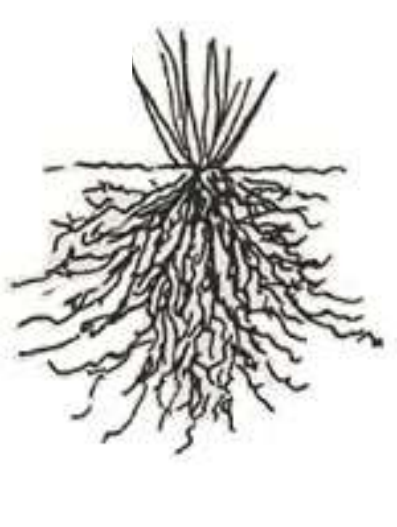
- Although we do not see much of roots, they are equally important to plant growth as stems and leaves …
  - Leaves collect light and CO_2 to produce sugar for growth as long as the plant can get enough water and nutrients from its roots!



Types of Root Systems

- A true root system consists of both primary & secondary roots (lateral roots)
- Primary root system:
  - Taproot System*
  - Fibrous Root System**
  - Adventitious Roots**





*Found in most **dicots** (flowering plants with 2 cotyledons in their seeds);

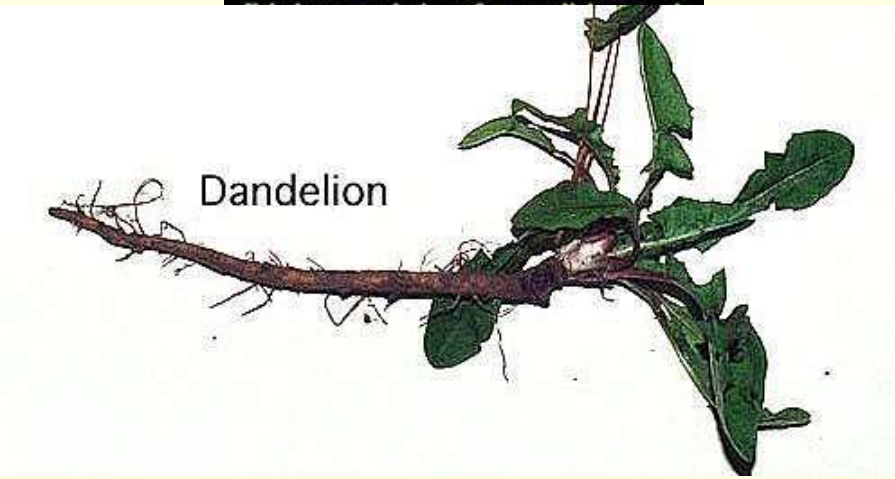
Example: Peas

Found in most **monocots (flowering plants with 1 cotyledon in their seeds);

Example: Corn

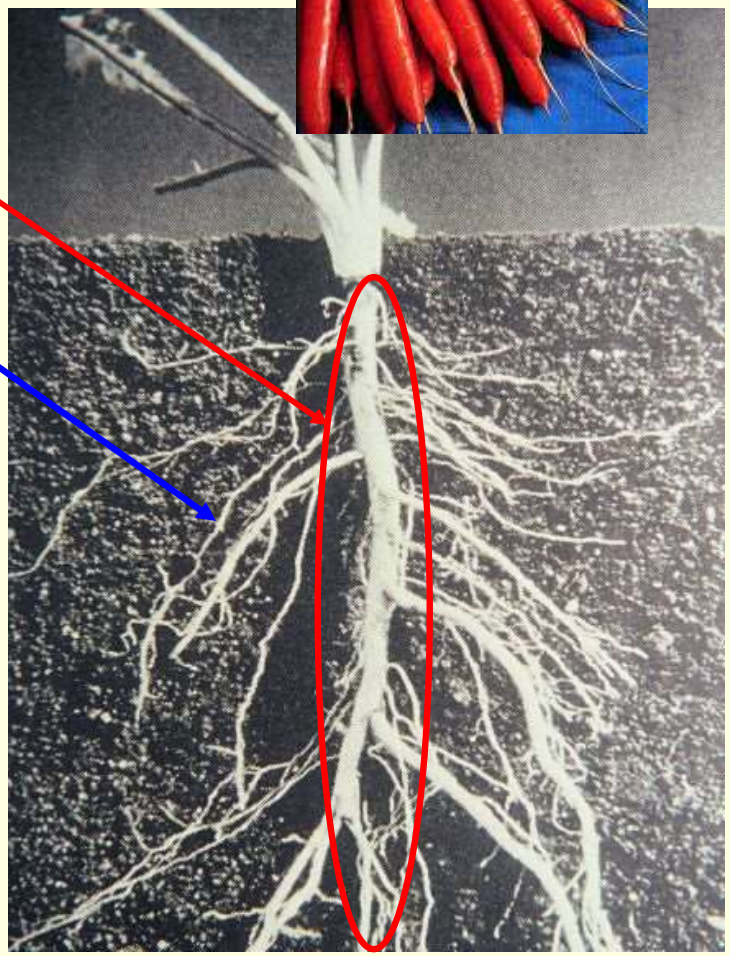
Primary Root System

- The 1st root to emerge from a seed is the **radicle** (primary root), which emerges to form a taproot that persists throughout the life of the plant.
- Many progressively smaller **branch roots (lateral roots)** grow from the taproot.



Taproot System

- Definition: the root system consisting of a thick, central primary root, which forms many smaller lateral roots.
  - Common in dicots and conifers.
- In some plants such as carrots and beets, fleshy taproots are the plant's "food pantry".
- In other plants such as poison ivy the long taproots are modified for reaching water deep in the ground.



Fibrous Root System

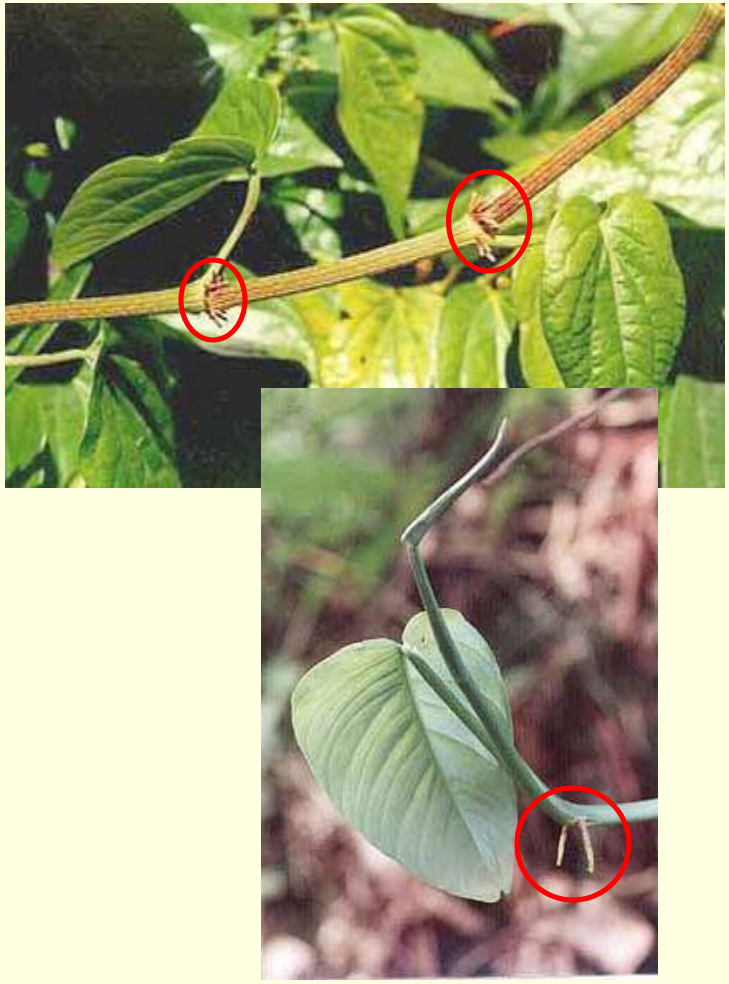
- Definition: root system that consists of several to many roots of the same size that develop from the end of the stem with smaller, lateral roots branching off of them.
- Typically slender, small roots
  - Found in monocots like onion, crabgrass, sweet potatoes, etc…
  - Form in plants in which the embryonic root is short-lived.

Fibrous Root System

- Most fibrous roots are very extensive and cling to soil particles, which is why they are often used for preventing erosion.
- Because fibrous roots do not arise from preexisting roots but rather from the stem, they are said to be **adventitious** …

Adventitious Roots

- Definition: a mass of roots that from on organs other than roots
- Usually begin growing after the seed germinates
- They form in all sorts of places on plants including leaves, petioles, and stems.



Adventitious Roots

Why Different Root Types?

- Help plants compete for water and minerals …

  Example: Prairie plants

  - Shallow, wide-spreading fibrous roots (<15 cm deep)
  - Medium-depth, evenly distributed fibrous roots (<1.5 m deep)
  - Deep taproots (lack width, but go very deep)

Root Functions

Primary & Specialized

Primary Root Functions

- Roots perform 4 main functions:
  - Absorption/Conduction of water and dissolved minerals
  - Anchorage
  - Storage (carbohydrates)
  - Production of hormones (e.g., gibberellins) & secondary plant compounds (e.g., nicotine in tobacco plants)

- Some specialized roots are modified for other functions ….

Specialized Roots

- Prop roots
- Aerial roots
- Strangling roots
- Contractile roots
- Parasitic roots (haustoria)
- Storage roots
- Buttress roots
- Pneumatophores

Prop or Stilt Roots

- Help to stabilize and support the plant.

  Examples:

  - Ficus
  - Corn

Aerial Roots

- Absorb water from the air; may also be photosynthetic

  Examples:

  - Orchids
  - Poison ivy

- Anchor plant to the soil and support the tree

  Example:

  - Banyan tree

Strangling Roots

- Help the plants to “climb” up another tree to get to sunlight; eventually will “strangle” the host plant (parasitic)

  Example: Fig

Contractile Roots

- Pull the plant shoot closer to the ground or deeper into the soil where the soil is relatively warm, helping plants survive variable early spring weather
  - Example: Dandelion

Parasitic Roots (Haustoria)

- Penetrate the stems & roots of host plant to obtain water, minerals, and organic molecules

  Examples:
  - Mistletoe
  - Devil's Dodder

Storage Roots

- Store water & food

  Examples:

  - Carrot
  - Radish
  - Beet
  - Turnip
  - Parsnip
  - Rutabaga

Buttress Roots

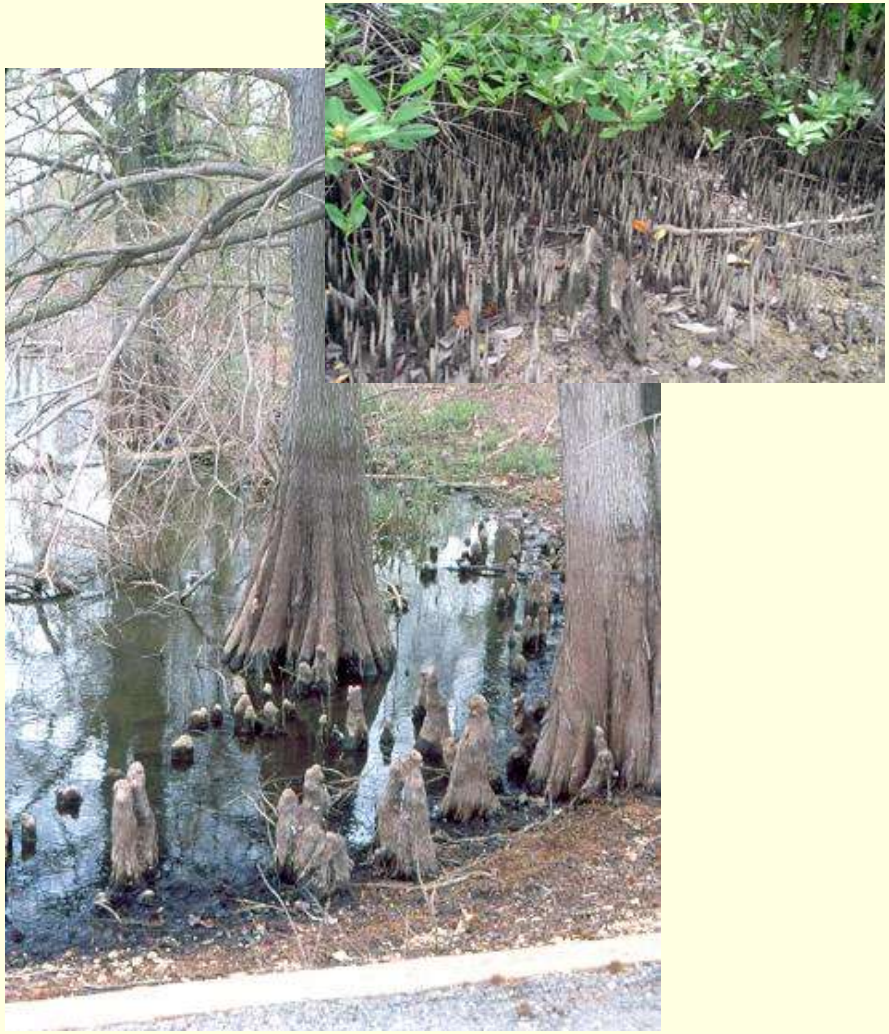
- Stabilize plants in thin soil & prevent uprooting
  - Common in tropical plants (e.g., ficus)

Pneumatophores (air roots)

- Supply oxygen to plants growing in swamps where water may be deoxygenated

  Examples:

  - Cypress trees
  - Mangrove trees



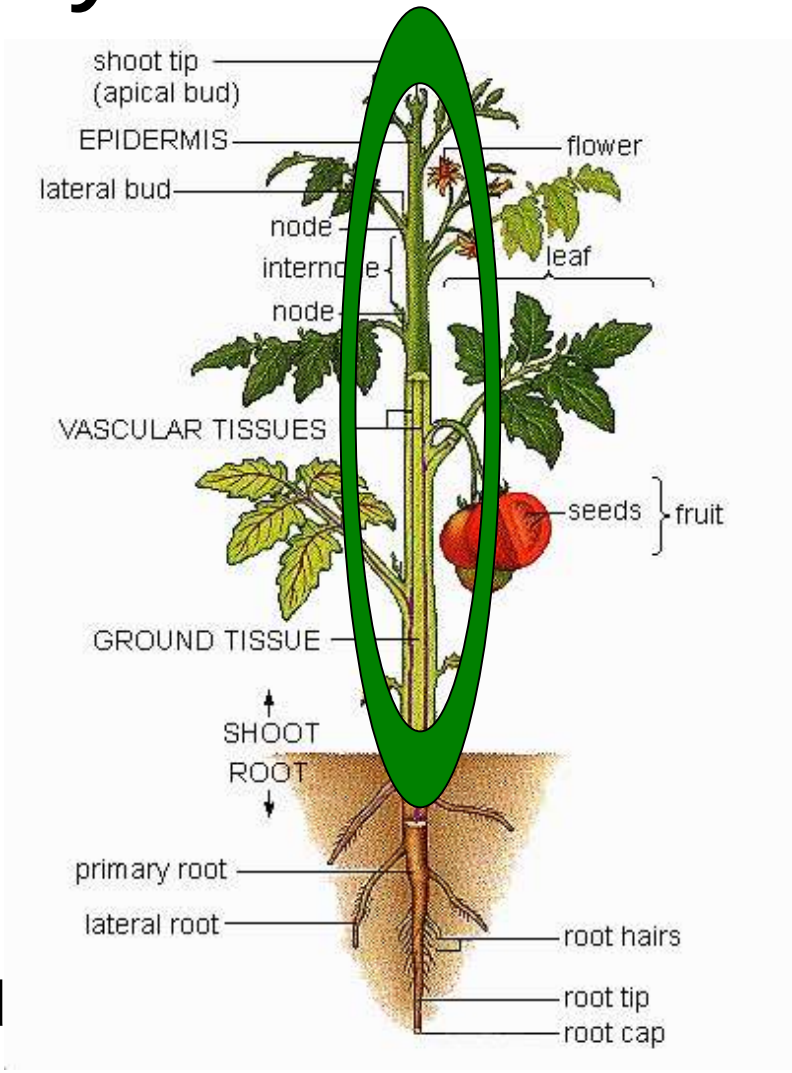
STEMS: FORM & FUNCTION

- Function
- External Anatomy
- Internal Anatomy
- Specialized Stems

The Plant Body: Stems

FUNCTION OF STEMS

- Stems support leaves and branches.
- Stems transport water and solutes between roots and leaves.
- Stems in some plants are photosynthetic.
- Stems may store materials necessary for life (e.g., water, starch, sugar).
- In some plants, stems have become adapted for specialized functions.



Stems support a display of leaves.

Stems orient the leaves toward the light with minimal overlap among the leaves.

Asclepias - milkweed

The stem supports a display of flowers

Cercis canadensis - redbud

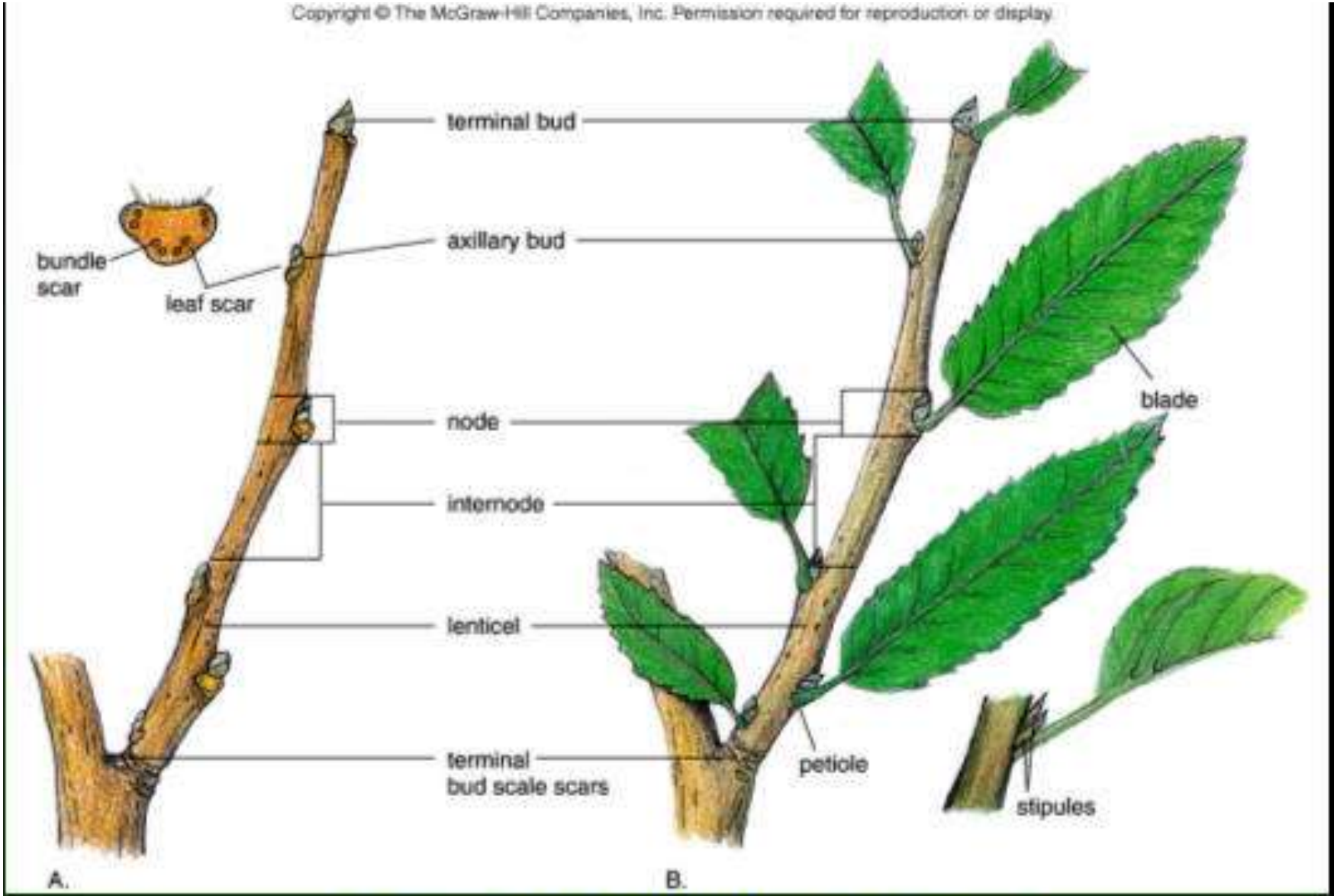
The stem does photosynthesis…and stores water.

Opuntia-prickly pear

This stem does photosynthesis, stores water, but also produces a defense chemical: mescaline…a hallucinogen.

Lophophora williamsii - peyote

EXTERNAL ANATOMY



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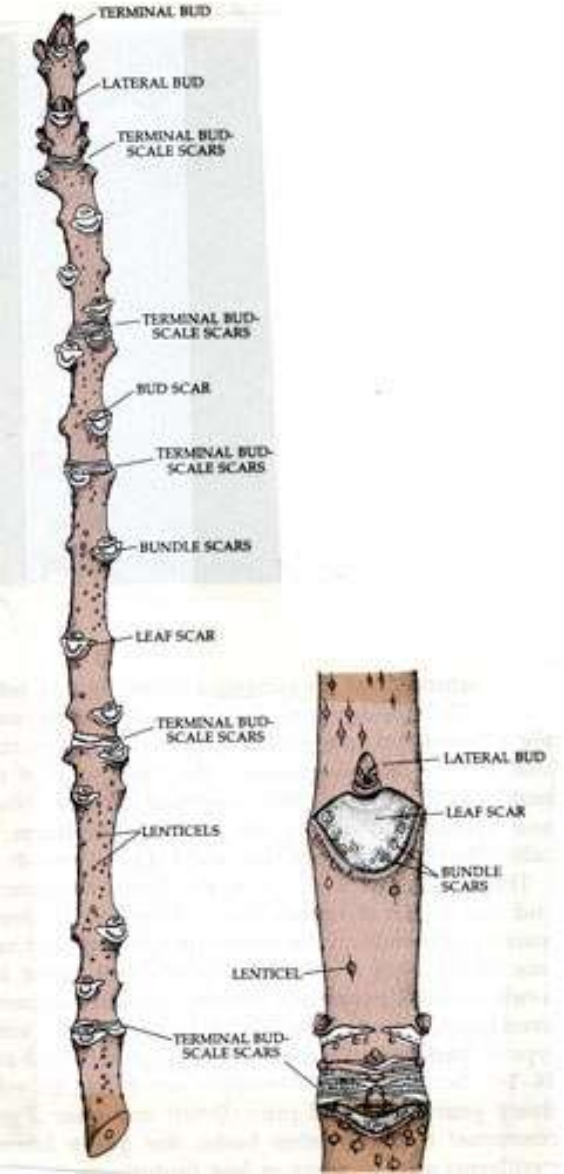
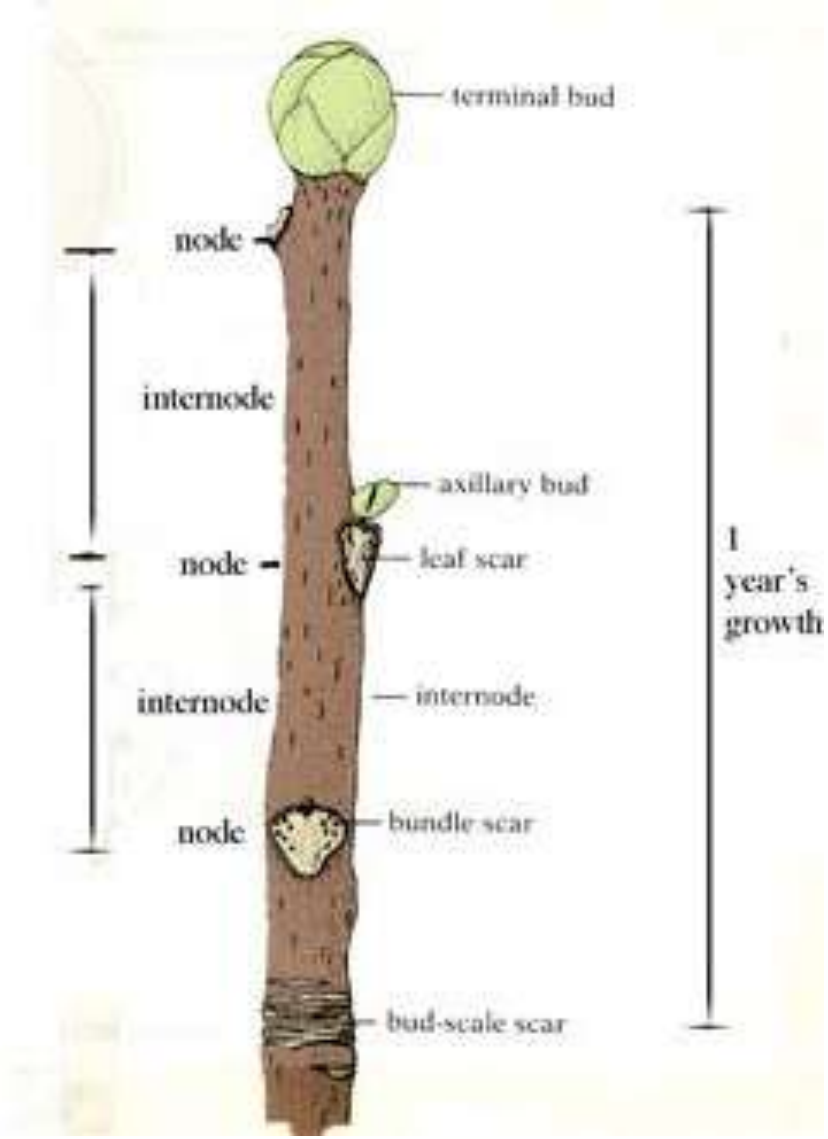
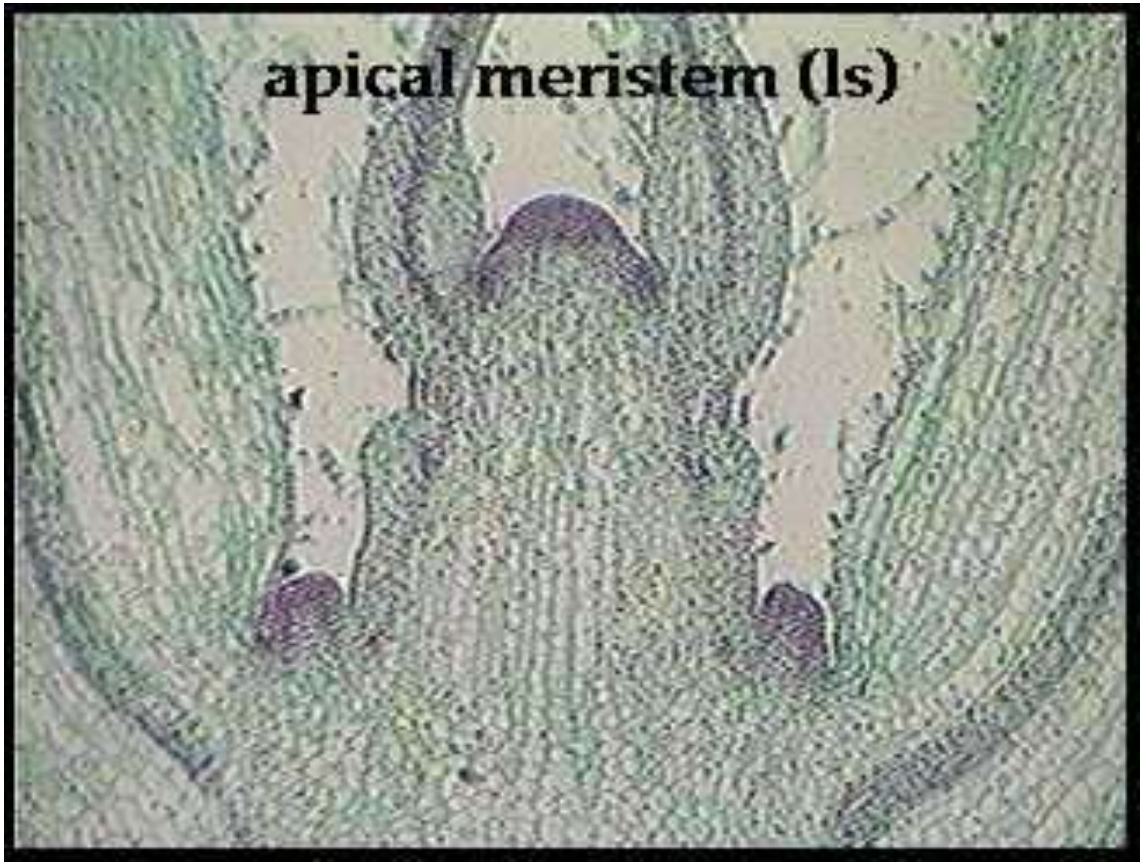
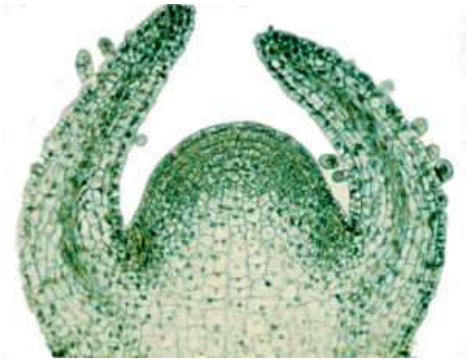
EXTERNAL ANATOMY

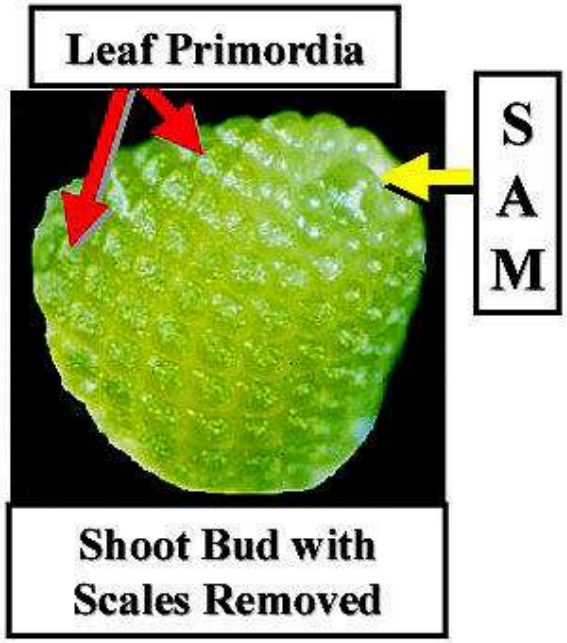




STEM APICAL MERISTEM







Apical Dominance

- ***Apical dominance*** refers to the suppression of growth by hormones produced in the apical meristem. The Christmas tree pattern of pines indicates strong apical dominance. Bushy plants have weak apical dominance. If apical meristem is eaten or destroyed, plants may become bushy.
  - Lateral branch growth are inhibited near the shoot apex, but less so farther from the tip.
  - Apical dominance is disrupted in some plants by removing the shoot tip, causing the plant to become bushy.

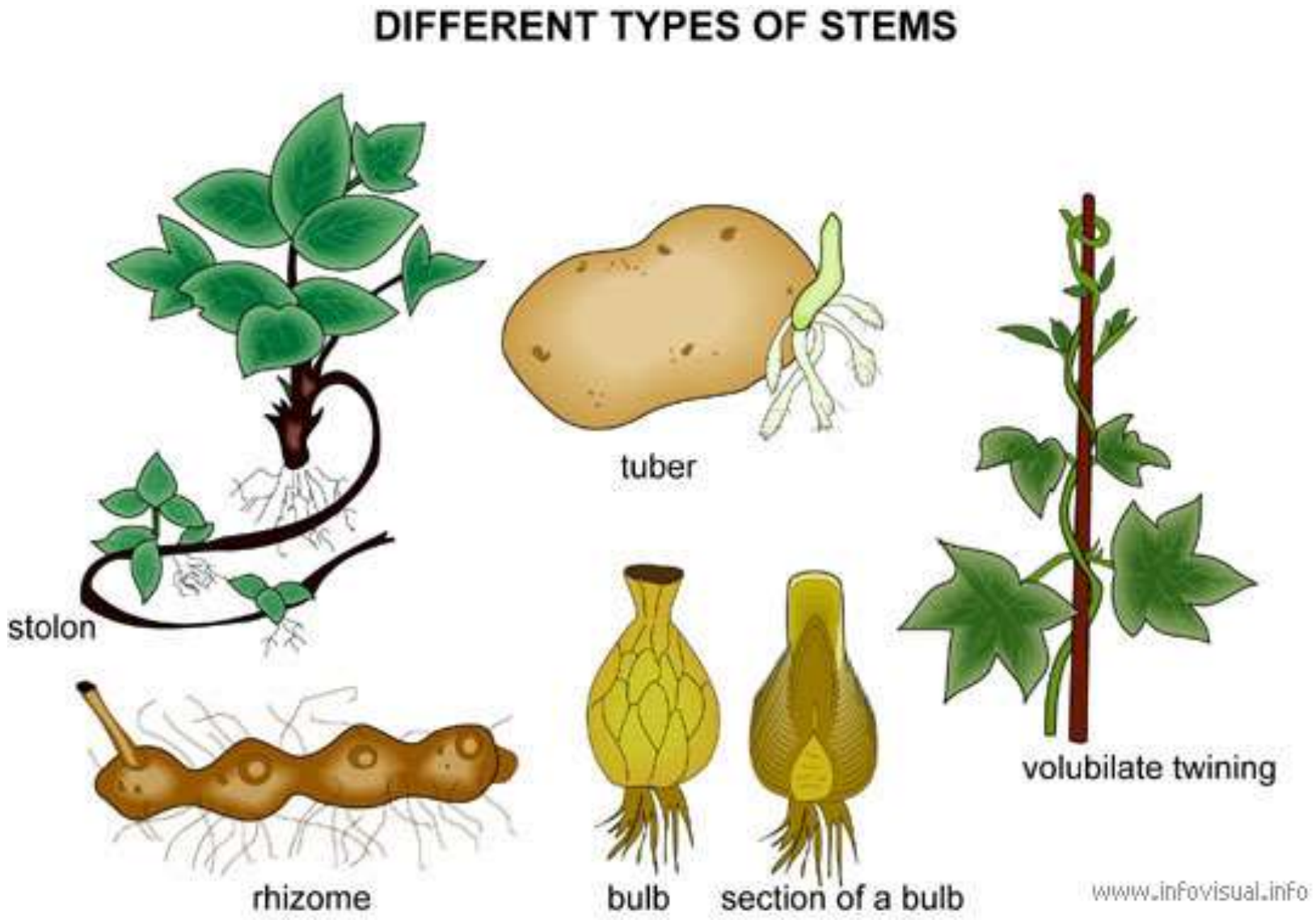
This tree is ***Pinus aristata*** (bristlecone pine).

One individual of this species shows more than 5000 growth rings!

Inner wood, harvested by boring, was used to validate carbon-14 dating.

Imagine the stories that this California tree could tell…perhaps something of migration of Asian peoples down the western coast of North America! They were contemporaries of Pharaohs!

Modified & Specialized Stems



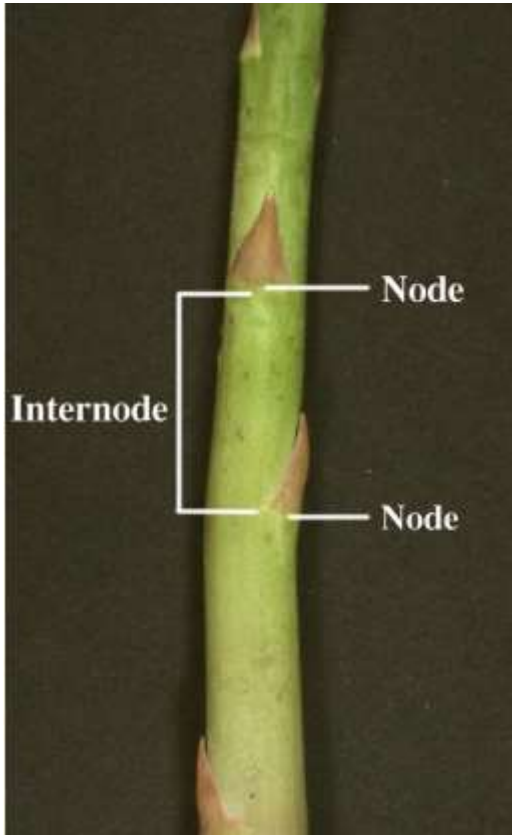
Food Storage Stems

Bamboo Shoots

Kohlrabi

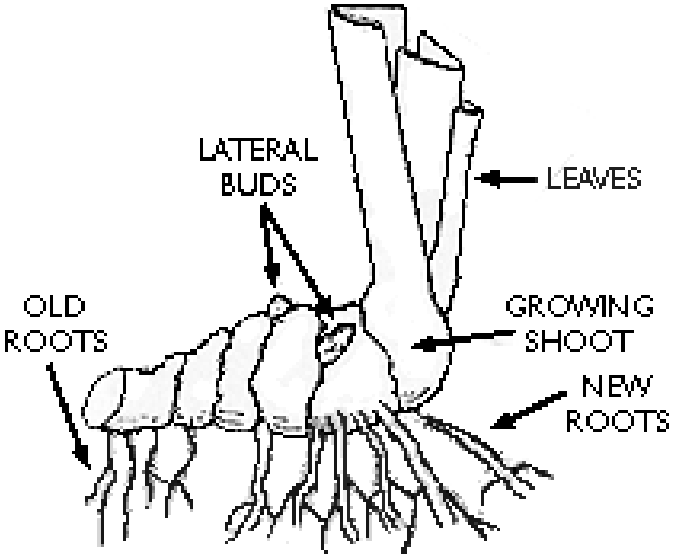
Food Storage Stems - Sugarcane

Food Storage Stems - Asparagus



Rhizomes

- Rhizomes - horizontal stems that grow below the ground with adventitious roots
- Examples of plants that can produce rhizomes are irises, ferns, and grasses.



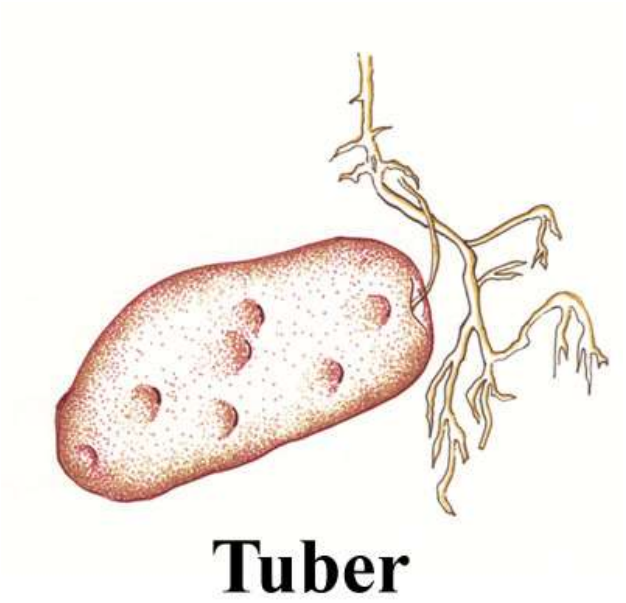
IRIS RHIZOME

Stolons

Stolon

- Stolons or runners - horizontal stem that grows above the ground with long internodes
- Examples of plants that can produce stolons are strawberry and airplane plants

Tuber



- Tubers - accumulation of food at the tips of underground stolons
- The "eyes" of a potato are the nodes of a starch-ladened stem

History of the Potato

Failure of the potato crop in 1845-49 led to the Irish Potato Famine

Taters and Spuds

Rosette

- Rosette - stem with short internodes and leaves attached at nodes

Wild Radish – Rosette & Bolt

A FLOWERING ANNUAL

YEAR ONE

YEAR ONE

Common Mullen – Rosette & Bolt

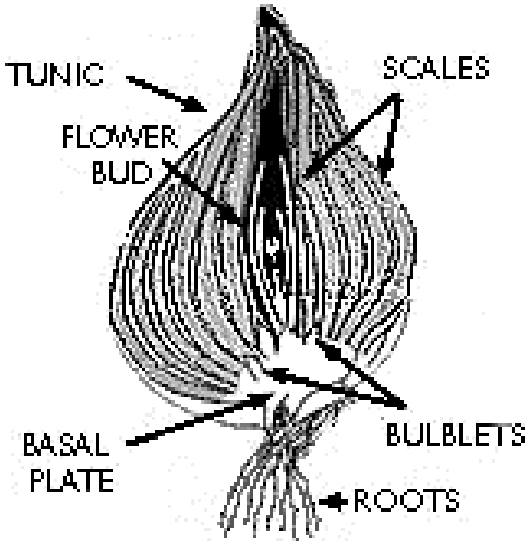
A FLOWERING BIENNIAL

YEAR ONE

YEAR TWO

Bulb

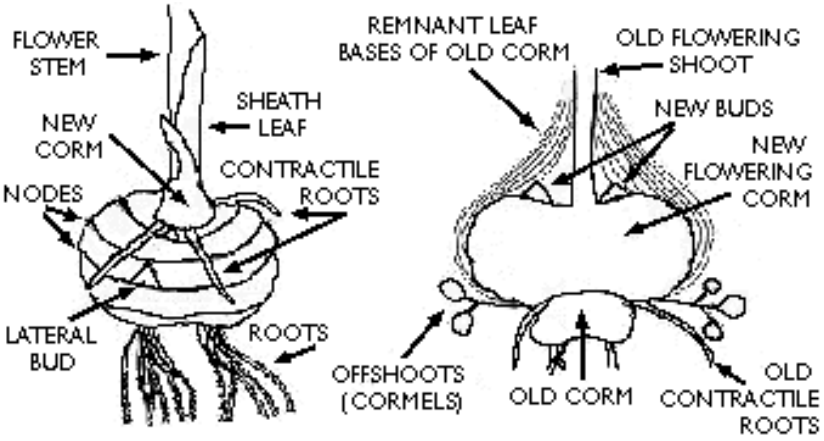
- Bulbs - large buds with a small stem at the lower end surrounded by numerous fleshy leaves, adventitious roots at base
- Examples include onion, tulip, and lily



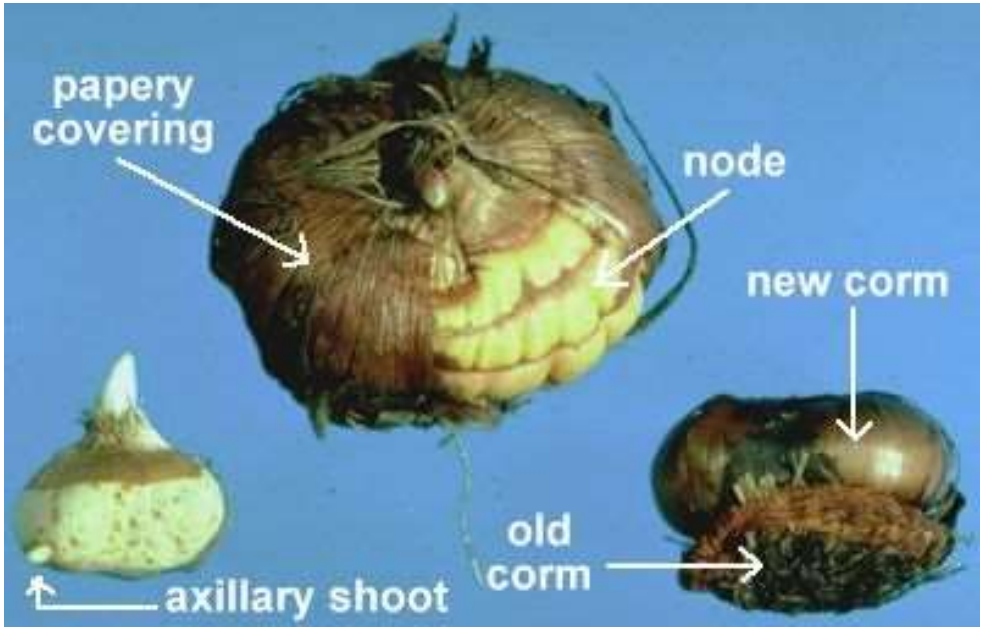
TULIP
(TUNICATE) BULB

Corm

- Corms - resemble bulbs but composed entirely of stem tissue surrounded by a few papery scale like leaves, food storage organs with adventitious roots at the base of corms
- Examples include crocus and gladiolus.



GLADIOLUS CORM



Cladophylls

- Cladophylls - leaf-like stems; examples include butcher's broom, asparagus

Photosynthetic Stems

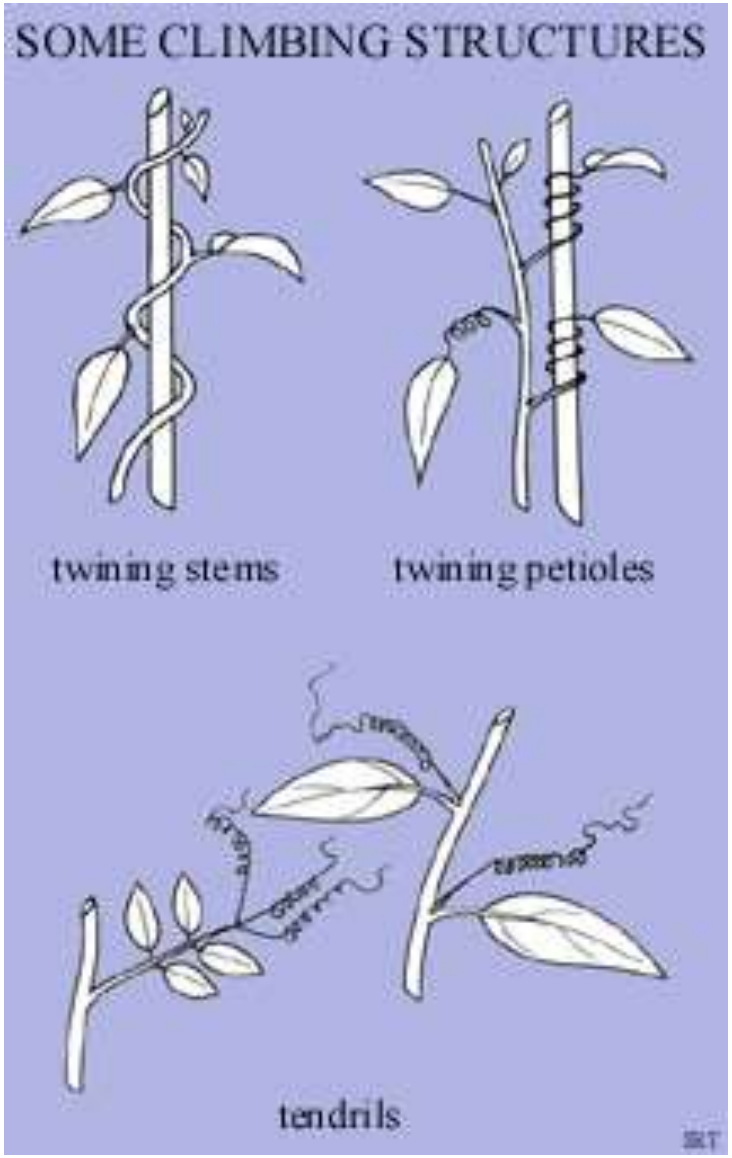
- Cacti - stout fleshy stems that are modified for food and water storage and photosynthesis.

Thorns

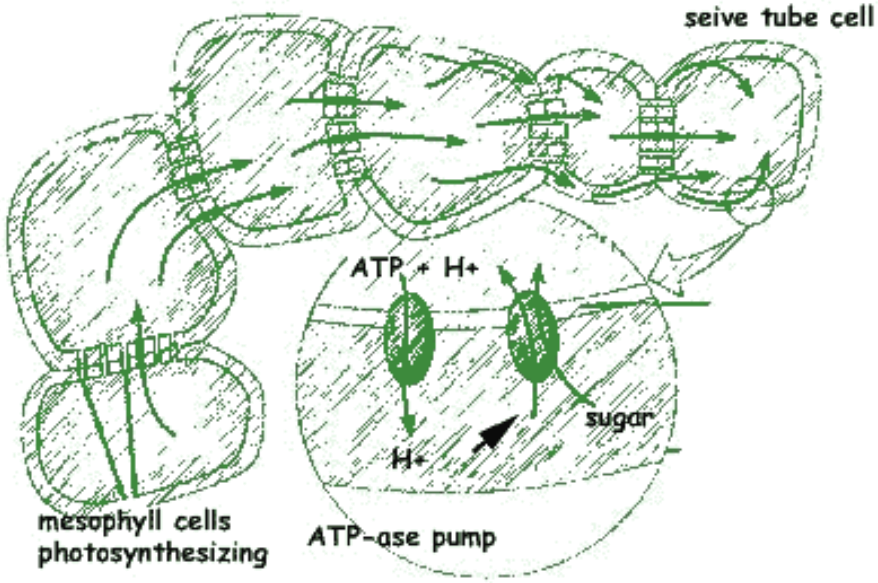
- Honey locust (modified stem)
- Black Locust (modified leaf stipules)

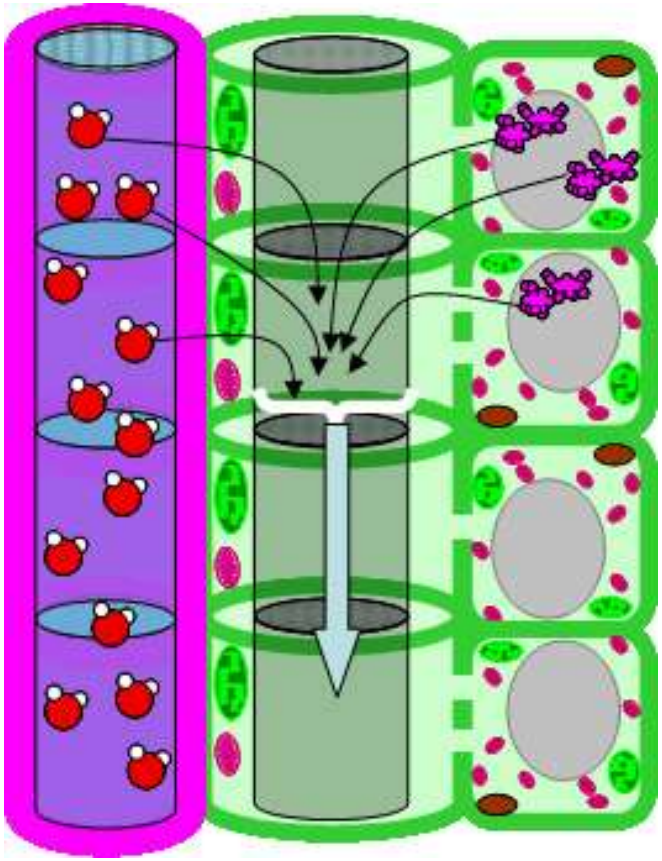
Tendrils

Grape Tendrils



Sugar Loading of Phloem and Bulk Flow





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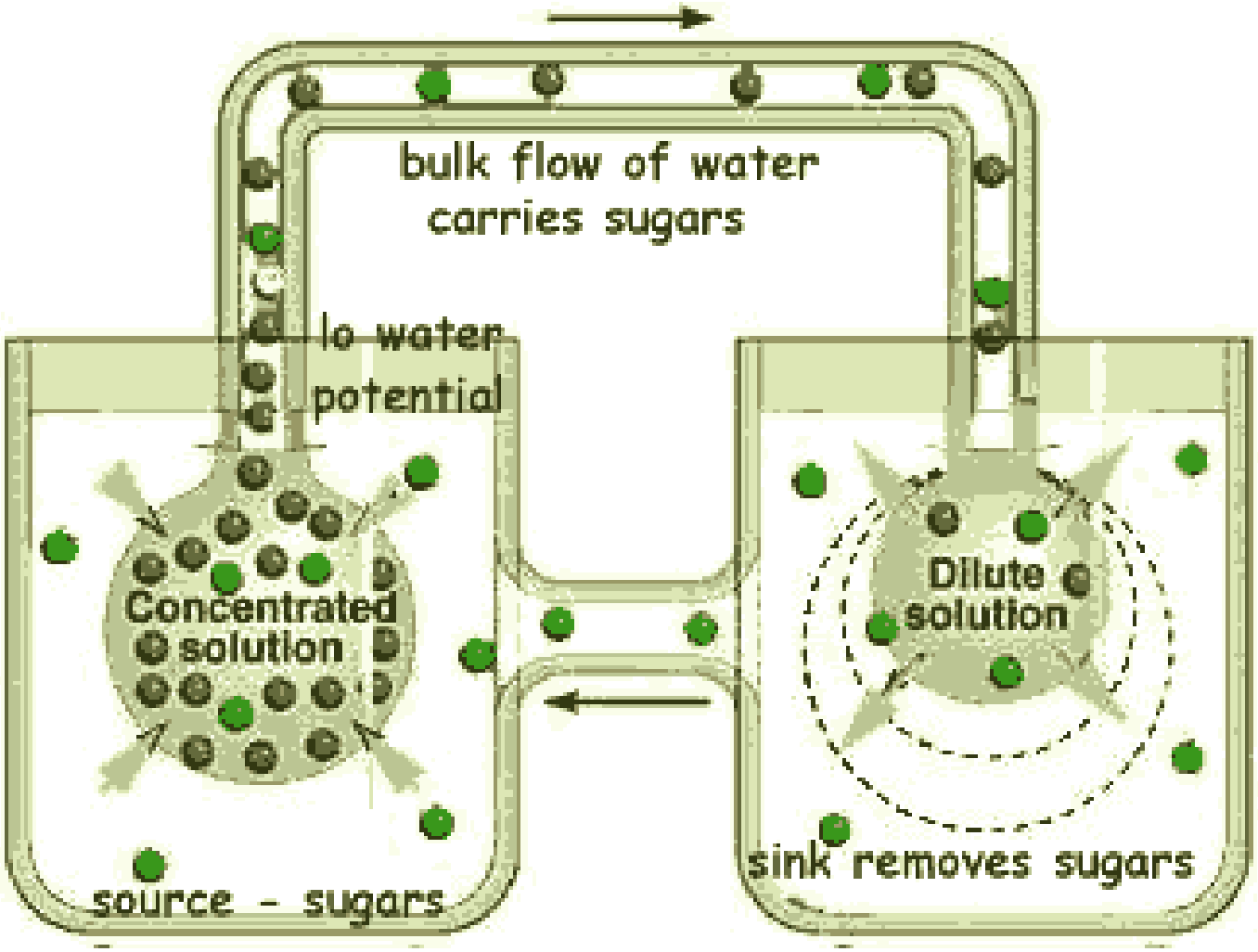
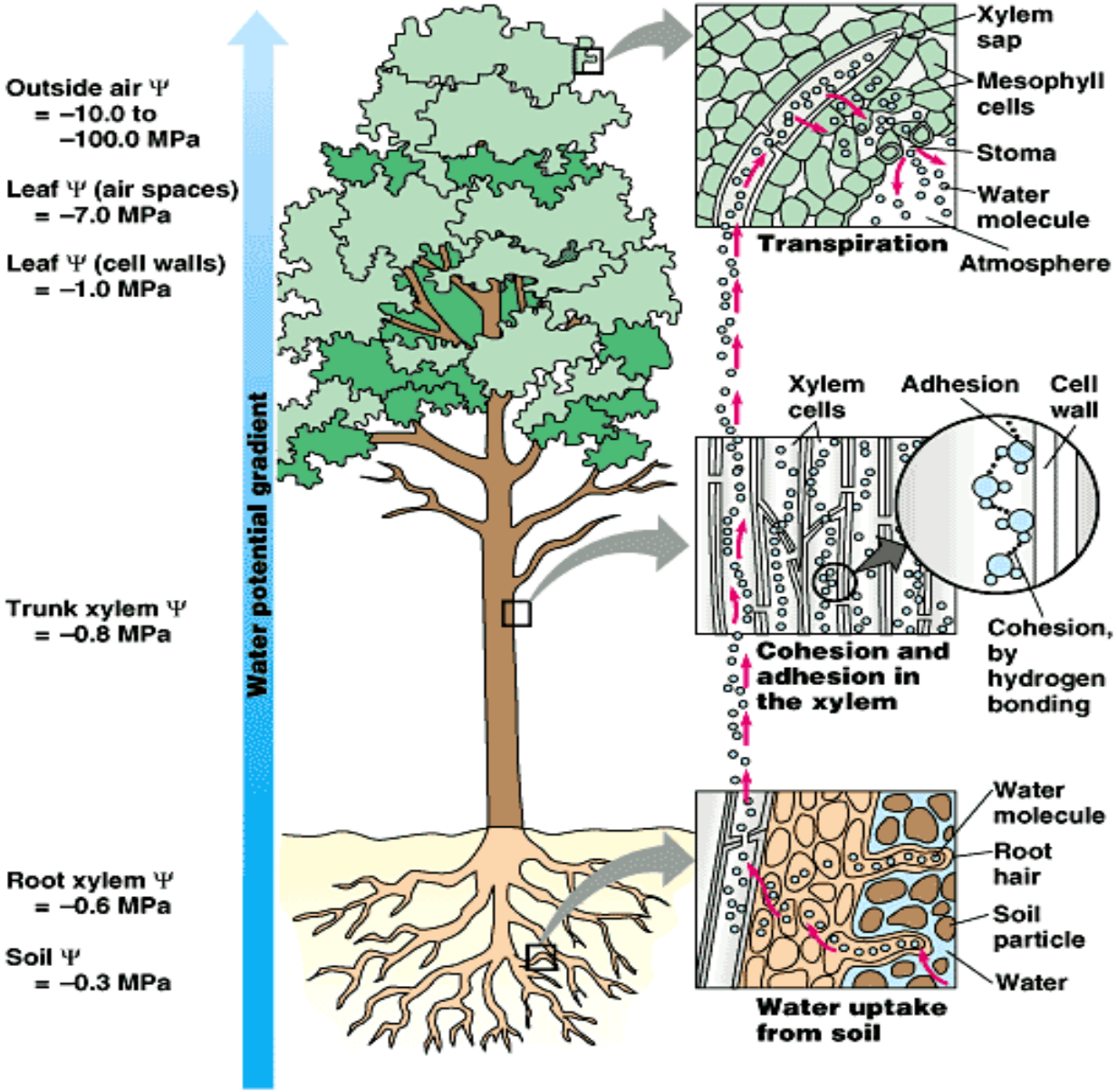


Transpiration-Cohesion Hypothesis



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